Final Project – Toronto's Neighbourhoods

Objective

For this project, you will use publicly available data from the City of Toronto's Open Data Portal to explore patterns and relationships across Toronto's neighbourhoods.

You will have the opportunity to develop your own research questions based on variables that interest you. Your analysis might focus on differences between neighbourhoods, relationships between neighbourhood characteristics, or patterns that could be relevant to Toronto residents and the city of Toronto. Your goal is to investigate an interesting question using data, not necessarily to find a strong relationship or a particular result.

Selected student projects may have the opportunity to be featured on the City of Toronto's Data stories!

Finding inspiration for your research question

One **OPTIONAL** way to develop a research question is to explore news articles and local stories about Toronto's neighbourhoods. You might find a story that raises an interesting question about housing, employment, crime, education, transportation, or another aspect of life in Toronto.

- Step 1) Try searching Google News or a local news website using keywords related to Toronto and a topic you're interested in.
    - You can explore City of Toronto's Data stories for examples of how data have been used to investigate questions about Toronto.
- Step 2) Identify something interesting in the story. As you read, consider:
    - Does the article describe a pattern or a relationship?
    - Does it make a claim that you could explore using the data?
    - Does it raise a question that you'd like to investigate further?
- Step 3) Connect the story to the dataset.
    - Identify variables that might help you explore the topic.

Remember: Your research question doesn't need to come directly from a news article. You are also welcome to explore the dataset and develop your own question based on relationships or patterns that interest you!

About the Dataset

The dataset combines information from several publicly available City of Toronto datasets. Because these datasets were collected at different times, not all variables represent the same year. When investigating relationships between variables, keep in mind that some neighbourhood characteristics may have changed over time.

The data set contains information about **158 Toronto neighbourhoods**, using data from the City of Toronto's Open Data.

Each row represents one Toronto neighbourhood, and each column represents a characteristic of that neighbourhood. Specifically, this dataset contains:

- 158 rows, for each of the Toronto neighbourhoods, and
- 80 variables/columns.

Note: Variables beginning with pct_ represent percentages on a scale from 0 to 100. For example, a value of 25 means 25% of the relevant population.

Data Dictionary

Neighbourhood information

These variables identify each neighbourhood and provide information about its classification. These data are based on the 2021 Census of Population data from Statistics Canada.

Variable	Definition
neighbourhood	Name of the Toronto neighbourhood.
neighbourhood_id	Unique identification number assigned to each neighbourhood.
designation	Classification of the neighbourhood as a Neighbourhood Improvement Area (NIA), Emerging Neighbourhood, or neither.
neighbourhood_area_km2	Total geographic area of the neighbourhood, measured in square kilometres (km^2)
toronto_region	Broad geographic region of Toronto in which the neighbourhood is located. Categories are Etobicoke York, North York, Scarborough, and Toronto and East York. Regions are based on the City of Toronto's Community Council boundaries.

Demographic information

These variables describe the characteristics of people living in each neighbourhood, including population size, age, household size, and immigration status. These data are based on the 2021 Census of Population data from Statistics Canada.

Variable	Definition
population	Total number of people living in the neighbourhood.
pct_children	Percentage of residents aged 0–14 years.
pct_seniors	Percentage of residents aged 65 years or older.
average_age	Average age of residents in the neighbourhood, in years.
median_age	Median age of residents in the neighbourhood, in years.
average_household_size	Average number of people living in a household.
pct_immigrants	Percentage of residents in private households who are immigrants.

Labour Force and Employment

These variables describe employment patterns among residents of each neighbourhood. These data are based on the 2021 Census of Population data from Statistics Canada.

The ***labour force*** includes people who are employed or unemployed AND actively looking for work. People who are not working and not actively looking for work are generally not considered part of the labour force.

Variable	Definition
participation_rate	Percentage of residents aged 15 or older who are in the labour force (either employed or unemployed and looking for work).
employment_rate	Percentage of residents aged 15 or older who are employed.
unemployment_rate	Percentage of people in the labour force who are unemployed.
pct_full_year_full_time_workers	Percentage of residents who worked in 2020 who worked full-time throughout the year.

pct_transit_commuters	Percentage of employed commuters who use public transport as their main mode of transportation to work, among those with a usual workplace or no fixed workplace address.
pct_walk_commuters	Percentage of employed commuters who walk as their main mode of transportation to work, among those with a usual workplace or no fixed workplace address.
pct_bike_commuters	Percentage of employed commuters who use a bicycle as their main mode of transportation to work, among those with a usual workplace or no fixed workplace address.

Note: The unemployment rate is calculated using only people in the labour force, whereas the employment rate is calculated using all residents aged 15 or older. These percentages have different denominators.

Education and Socioeconomic Characteristics

These variables describe educational attainment, income, and economic conditions across Toronto's neighbourhoods. These data are based on the 2021 Census of Population data from Statistics Canada.

Variable	Definition
pct_no_certificate_25_64	Percentage of residents aged 25–64 with no educational certificate, diploma, or degree.
pct_high_school_only_25_64	Percentage of residents aged 25–64 whose highest qualification is a high school diploma or equivalent.
pct_postsecondary_belowbach_25_64	Percentage of residents aged 25–64 whose highest qualification is a postsecondary certificate or diploma below the bachelor's degree level, including trades, college, and certain university certificates.
pct_bachelors_only_25_64	Percentage of residents aged 25–64 whose highest qualification is a bachelor's degree (excluding master's and doctoral degrees).
pct_other_abovebach_25_64	Percentage of residents aged 25–64 whose highest qualification is a university certificate or diploma above the bachelor's degree level, excluding master's degrees, medicine degrees, and doctorates.
pct_deg_medicine_25_64	Percentage of residents aged 25–64 whose highest qualification is a degree in medicine, dentistry, veterinary medicine, or optometry.
pct_masters_25_64	Percentage of residents aged 25–64 whose highest qualification is a master's degree (excluding doctoral degrees).
pct_doctorate_25_64	Percentage of residents aged 25–64 whose highest qualification is an earned doctorate (e.g., PhD).
median_household_income	Median total annual household income in 2020, measured in Canadian dollars. Half of households have incomes below this value, and approximately half have incomes above it.
low_income_pct	Percentage of residents classified as having low income according to the after-tax Low-income Measure (LIM-AT). This measure identifies individuals whose adjusted household income falls below a specified low-income threshold.

Housing and Cost of Living

These variables describe housing costs, affordability, and housing characteristics across Toronto's neighbourhoods. They include information about renting, home values, housing conditions, and the types of dwellings found in each neighbourhood.

Variable	Definition
pct_renters	Percentage of private households that rent their homes rather than own them.

pct_highrise	Percentage of occupied private dwellings that are apartments in buildings with five or more storeys.
median_monthly_rent	Median monthly shelter costs for rented dwellings, measured in Canadian dollars. Includes rent and applicable utilities and municipal services.
median_home_value	Median estimated value of owner-occupied dwellings, measured in Canadian dollars.
pct_renters_30plus_shelter	Percentage of tenant households spending 30% or more of their household income on shelter costs.
pct_owners_30plus_shelter	Percentage of owner households spending 30% or more of their household income on shelter costs.
pct_subsidized_renters	Percentage of tenant households living in subsidized housing, where rent is reduced through government or other assistance programs.
pct_condos	Percentage of occupied private dwellings that are condominiums.
pct_major_repairs	Percentage of occupied private dwellings requiring major repairs, such as repairs to plumbing, electrical systems, or structural components.
pct_older_housing	Percentage of occupied private dwellings built in 1980 or earlier.
majority_renters	Indicates whether more than 50% of private households in the neighbourhood rent their homes. Categories: Yes or No.
majority_highrise	Indicates whether more than 50% of occupied private dwellings in the neighbourhood are apartments in buildings with five or more storeys. Categories: Yes or No.

Crime and Public safety

These variables describe rates of different reported crimes across Toronto neighbourhoods in 2025.

How are the crime rates calculated? Crime rates represent the number of reported crimes per 100,000 residents in a neighbourhood over one year. They are calculated as:

$$\text{Crime Rate} = \frac{\text{Number of reported crimes}}{\text{Neighbourhood population}} \times 100{,}000$$

For example, if a neighbourhood has a robbery rate of 200, this means that this neighbourhood had 200 reported robberies per 100,000 residents that year.

Variable	Definition
assault_rate_2025	Rate of reported assaults in the neighbourhood in 2025. Assault involves the use or threat of physical force against another person.
autotheft_rate_2025	Rate of reported motor vehicle thefts in the neighbourhood in 2025.
biketheft_rate_2025	Rate of reported bicycle thefts in the neighbourhood in 2025.
breakenter_rate_2025	Rate of reported break-and-enter incidents in the neighbourhood in 2025, involving unlawful entry into a building with intent to commit an offence.
homicide_rate_2025	Rate of reported homicides in the neighbourhood in 2025.
robbery_rate_2025	Rate of reported robberies in the neighbourhood in 2025. Robbery involves taking or attempting to take property using force or threats.
shooting_rate_2025	Rate of reported shooting incidents in the neighbourhood in 2025.
theftfrommv_rate_2025	Rate of reported thefts of property from motor vehicles in the neighbourhood in 2025.

Note: Some neighbourhoods have missing values (NA) for bicycle theft, homicide, and shooting rates. These values are also missing from the corresponding crime counts in the original dataset. They have been retained as missing values rather than assumed to be zero. You can still use these variables in your analysis, but R will generally exclude observations with missing values when creating plots or calculating statistics. Keep in mind that your analysis may therefore include fewer than 158 neighbourhoods.

Mental Health and Crisis Response

These variables describe calls for service attended by Toronto Police Service officers involving persons in crisis, suicide-related incidents, and overdoses in 2025. All variables represent the number of police-attended calls per 100,000 neighbourhood residents.

Variable	Definition
crisis_rate_2025	Rate of police-attended Person in Crisis calls per 100,000 residents in 2025.
suicide_related_rate_2025	Rate of police-attended calls involving attempted or threatened suicide per 100,000 residents in 2025.
overdose_rate_2025	Rate of police-attended overdose calls per 100,000 residents in 2025.
total_pic_rate_2025	Combined rate of all three types of police-attended calls per 100,000 residents in 2025.

Traffic Enforcement

These variables describe the number of traffic tickets issued in Toronto's neighbourhoods in 2025, grouped by the type of traffic offence.

Variable	Definition
speeding_tickets_2025	Number of speeding tickets issued in the neighbourhood in 2025.
aggressive_driving_tickets_2025	Number of tickets issued for offences classified as aggressive driving in the neighbourhood in 2025.
distracted_driving_tickets_2025	Number of tickets issued for offences classified as distracted driving in the neighbourhood in 2025.
moving_violation_tickets_2025	Number of tickets issued for offences classified as moving violations in the neighbourhood in 2025.

Parks, green spaces, and recreation

These variables describe parks, green spaces, and recreational facilities across Toronto neighbourhoods.

The park data are based on geographic areas classified as "Park" in the City of Toronto's Green Spaces dataset. Other green spaces, such as hydro corridors, cemeteries, and certain conservation areas, are excluded.

These data were prepared by your instructors using boundary files published on the open data portal.

Variable	Definition
num_parks	Number of distinct named parks located at least partially within the neighbourhood. Parks that extend across multiple neighbourhoods are counted in each neighbourhood they overlap.
park_area_km2	Total area of land classified as parkland within the neighbourhood, measured in square kilometres (km^2).
pct_parkland	Percentage of the neighbourhood's total geographic area classified as parkland.

num_tennis_courts	Number of tennis courts located within the neighbourhood, including public courts and courts operated by tennis clubs.
has_tennis_courts	Indicates whether the neighbourhood has at least one tennis court identified in the City of Toronto's Tennis Court facilities dataset. Categories: Yes or No.
num_playgrounds	Number of playground facilities located within the neighbourhood.
num_pickleball_courts	Number of facilities classified as pickleball courts located within the neighbourhood.
num_sports_fields	Number of facilities classified as sports fields located within the neighbourhood.
num_ball_diamonds	Number of baseball or softball diamonds located within the neighbourhood.
num_splash_pads	Number of splash pad facilities located within the neighbourhood.
num_basketball_courts	Number of outdoor basketball court facilities located within the neighbourhood.
num_community_gardens	Number of community garden facilities located within the neighbourhood.

Businesses and Restaurants

These variables describe the number and concentration of selected types of licenced businesses across Toronto's neighbourhoods. The data were prepared by your instructors using the City of Toronto's Business Licenses and Address Points datasets.

How was business density calculated? The number of businesses within a geographical area. It is calculated by dividing the number of businesses by the neighbourhood's area in square kilometres.

$$\text{Business Density} = \frac{\text{Number of Businesses}}{\text{Neighbourhood Area (km}^2\text{)}}$$

Variable	Definition
num_eating_establishments	Number of licensed eating or drinking establishments in the neighbourhood, including restaurants, bars, and other establishments serving food or beverages.
num_takeout_food	Number of licensed take-out or retail food establishments in the neighbourhood.
num_personal_services	Number of licensed personal service establishments in the neighbourhood, such as hair salons, barbershops, and beauty salons.
num_public_garages	Number of establishments licensed as public garages, including certain automotive service and repair businesses.
num_sidewalk_cafes	Number of licensed sidewalk cafés in the neighbourhood. A sidewalk café is an outdoor seating area associated with a restaurant, café, or bar that extends to the public sidewalk.
num_laundromats	Number of licensed laundry establishments in the neighbourhood.
num_entertainment_venues	Number of licensed entertainment establishments in the neighbourhood, including nightclubs, amusement establishments, and other entertainment venues.
eating_establishments_per_km2	Number of licensed eating or drinking establishments per square kilometre of neighbourhood area.
takeout_food_per_km2	Number of licensed take-out or retail food establishments per square kilometre of neighbourhood area.
has_entertainment	Indicates whether at least one licensed entertainment establishment was identified in the neighbourhood. Categories: Yes or No.

has_sidewalk_cafe	Indicates whether at least once licensed sidewalk café was identified in the neighbourhood. A sidewalk café is an outdoor seating area associated with a restaurant, café, or bar that extends to the public sidewalk. Categories: Yes or No.

Note: These variables are based on City of Toronto business licence records. They include businesses whose licences have not been cancelled and whose addresses could be matched to a neighbourhood. Although the counts may not include every operating business, they can be used to explore patterns and differences in licensed business activity across Toronto's neighbourhoods.